

eda

July 14, 2026

1 Orbita Digital - EDA Ventas

```
[53]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

1.0.1 Load Data

```
[54]: wb = pd.ExcelFile("../data/raw/Ventas_OrbitaDigital.xlsx")
wb.sheet_names
```

```
[54]: ['Hoja1']
```

```
[55]: df_ventas = pd.read_excel("../data/raw/Ventas_OrbitaDigital.xlsx",
↪sheet_name="Hoja1")
df_ventas.head()
```

```
[55]:
```

	Fecha De Venta	Ubicación	
0	2020-06-01	Brasil, América del Sur	
1	2020-06-01	Estados Unidos, América del Norte	
2	2020-06-01	Eslovenia, Europa	
3	2020-06-01	Canadá, América del Norte	
4	NaT	NaN	

	Producto	Marca	Categoría	Precio Unidad	
0	Juego Mesa for kids E300 Prata	Kid Toys	Juegos de mesa	5.5	
1	Juego Mesa for kids E300 Prata	Kid Toys	Juegos de mesa	5.5	
2	Juego Mesa for kids E300 Prata	Kid Toys	Juegos de mesa	5.5	
3	Juego Mesa for kids E300 Prata	Kid Toys	Juegos de mesa	5.5	
4	NaN	NaN	NaN	NaN	

	Costo Unidad	Cantidad Vendida	Facturación
0	2.8	1.0	5.5
1	2.8	1.0	5.5
2	2.8	2.0	11.0
3	2.8	1.0	5.5
4	NaN	NaN	NaN

1.0.2 Data Overview

This section provides a general summary of dataset before performing other sections such as data quality assesment, exploratory data visualization, etc.

```
[57]: overview = pd.DataFrame({
    "Metric": ["Number of records", "Number of columns", "Time period",
    ↪ "Categories",
    "Locations", "Products", "Brands"],
    "Values": [
        df_ventas.shape[0],
        df_ventas.shape[1],
        f"{df_ventas['Fecha De Venta'].min().strftime('%b %Y')} -
    ↪ {df_ventas['Fecha De Venta'].max().strftime('%b %Y')}",
        df_ventas['Categoría'].nunique(),
        df_ventas['Ubicación'].nunique(),
        df_ventas['Producto'].nunique(),
        df_ventas['Marca'].nunique(),
    ]
})

overview
```

```
[57]:
```

	Metric	Values
0	Number of records	22419
1	Number of columns	9
2	Time period	Jun 2020 - Aug 2022
3	Categories	20
4	Locations	37
5	Products	444
6	Brands	7

Data types

```
[61]: data_types = pd.DataFrame({
    "Column": df_ventas.columns,
    "Data Type": df_ventas.dtypes.values
})

data_types
```

```
[61]:
```

	Column	Data Type
0	Fecha De Venta	datetime64[us]
1	Ubicación	str
2	Producto	str
3	Marca	str
4	Categoría	str
5	Precio Unidad	float64
6	Costo Unidad	float64

7	Cantidad Vendida	float64
8	Facturación	float64

Observations

- The dataset contains historical information about sales transactions
- Apparently each row represents a different sale transaction, since there is not a identifier for each sale.
- The dataset has more than 22,000 records and 9 features or columns
- There are 37 different locations, but in the same column are the country and continent
- The 444 different products belong to 7 brands and 20 categories.

1.0.3 Dataset Quality Assesment

```
[63]: overview_quality = pd.DataFrame({
      "Metric": ["Missing Values", "Duplicated"],
      "Values": [
          df_ventas.isna().sum().sum(),
          df_ventas.duplicated().sum(),
      ]
    })

overview_quality
```

```
[63]:
```

	Metric	Values
0	Missing Values	90
1	Duplicated	2759

Where are the missing values?

```
[71]: na_df = pd.DataFrame({
      "Column": df_ventas.columns,
      "Missing Values": df_ventas.isna().sum().values
    })

na_df
```

```
[71]:
```

	Column	Missing Values
0	Fecha De Venta	10
1	Ubicación	10
2	Producto	10
3	Marca	10
4	Categoría	10
5	Precio Unidad	10
6	Costo Unidad	10
7	Cantidad Vendida	10
8	Facturación	10

Where is duplicated data

```
[67]: df_ventas[df_ventas.duplicated()].head(10)
```

```
[67]:
```

	Fecha De Venta	Ubicación \
10	NaT	NaN
16	NaT	NaN
17	NaT	NaN
24	2020-06-01	Estados Unidos, América del Norte
25	2020-06-01	Estados Unidos, América del Norte
27	2020-06-01	Estados Unidos, América del Norte
28	2020-06-01	Brasil, América del Sur
29	2020-06-01	Estados Unidos, América del Norte
30	2020-06-01	Eslovenia, Europa
31	2020-06-01	Canadá, América del Norte

	Producto	Marca	Categoría	Precio	Unidad \
10	NaN	NaN	NaN		NaN
16	NaN	NaN	NaN		NaN
17	NaN	NaN	NaN		NaN
24	Age of Mythology 2009 E158	Kid Toys	Juegos		32.0
25	Age of Mythology 2009 E158	Kid Toys	Juegos		32.0
27	Juego Mesa for kids E300 Prata	Kid Toys	Juegos de mesa		5.5
28	Juego Mesa for kids E300 Prata	Kid Toys	Juegos de mesa		5.5
29	Juego Mesa for kids E300 Prata	Kid Toys	Juegos de mesa		5.5
30	Juego Mesa for kids E300 Prata	Kid Toys	Juegos de mesa		5.5
31	Juego Mesa for kids E300 Prata	Kid Toys	Juegos de mesa		5.5

	Costo Unidad	Cantidad Vendida	Facturación
10	NaN	NaN	NaN
16	NaN	NaN	NaN
17	NaN	NaN	NaN
24	16.31	1.0	32.0
25	16.31	1.0	32.0
27	2.80	1.0	5.5
28	2.80	1.0	5.5
29	2.80	1.0	5.5
30	2.80	2.0	11.0
31	2.80	1.0	5.5

Unique Values

```
[68]: unique_data = pd.DataFrame({
    "Column": ["Ubicación", "Producto", "Categoría", "Marca"],
    "Unique Values": [
        df_ventas["Ubicación"].unique(),
        df_ventas["Producto"].unique(),
        df_ventas["Categoría"].unique(),
        df_ventas["Marca"].unique()
    ]
})
```

```
}
unique_data
```

```
[68]:
```

	Column	Unique Values
0	Ubicación	[Brasil, América del Sur, Estados Unidos, Amér...
1	Producto	[Juego Mesa for kids E300 Prata, nan, Juego Me...
2	Categoría	[Juegos de mesa, nan, Juegos, Auriculares Blue...
3	Marca	[Kid Toys, nan, Philips, Ninja, Sonos, Sony, L...

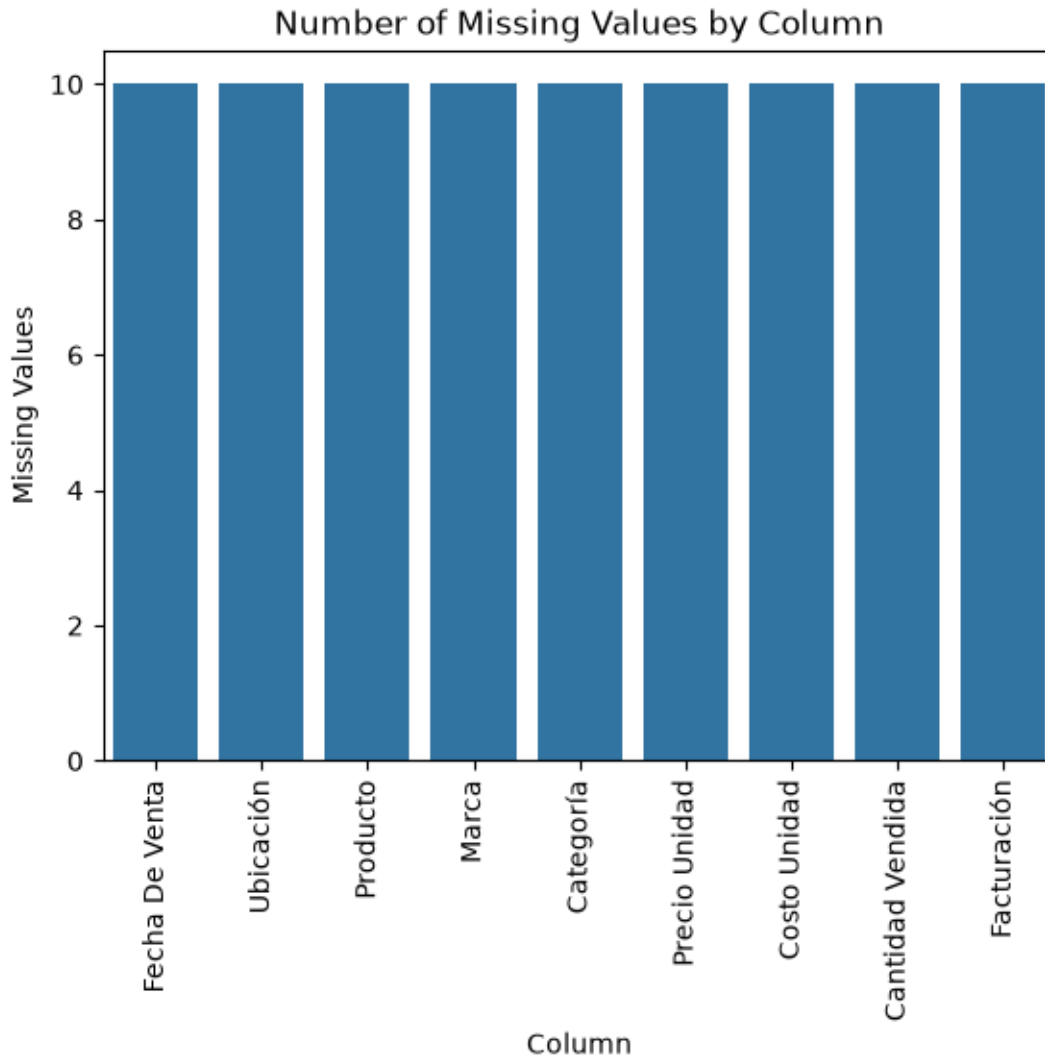
Observations

- The dataset has 90 missing values and more than 2,000 duplicated data
- Each column contains 10 missing values
- Duplicated data apparently represents a single sale transaction.
- It is necessary drop missing values because entire rows have missing values, and it is not necessary drop duplicated data.
- The Ubicación column must be divided into País and Continent

1.0.4 Exploratory Data Visualization

Missing Values

```
[74]: sns.barplot(data= na_df, x="Column", y="Missing Values")
plt.title("Number of Missing Values by Column")
plt.xticks(rotation=90)
plt.show()
```



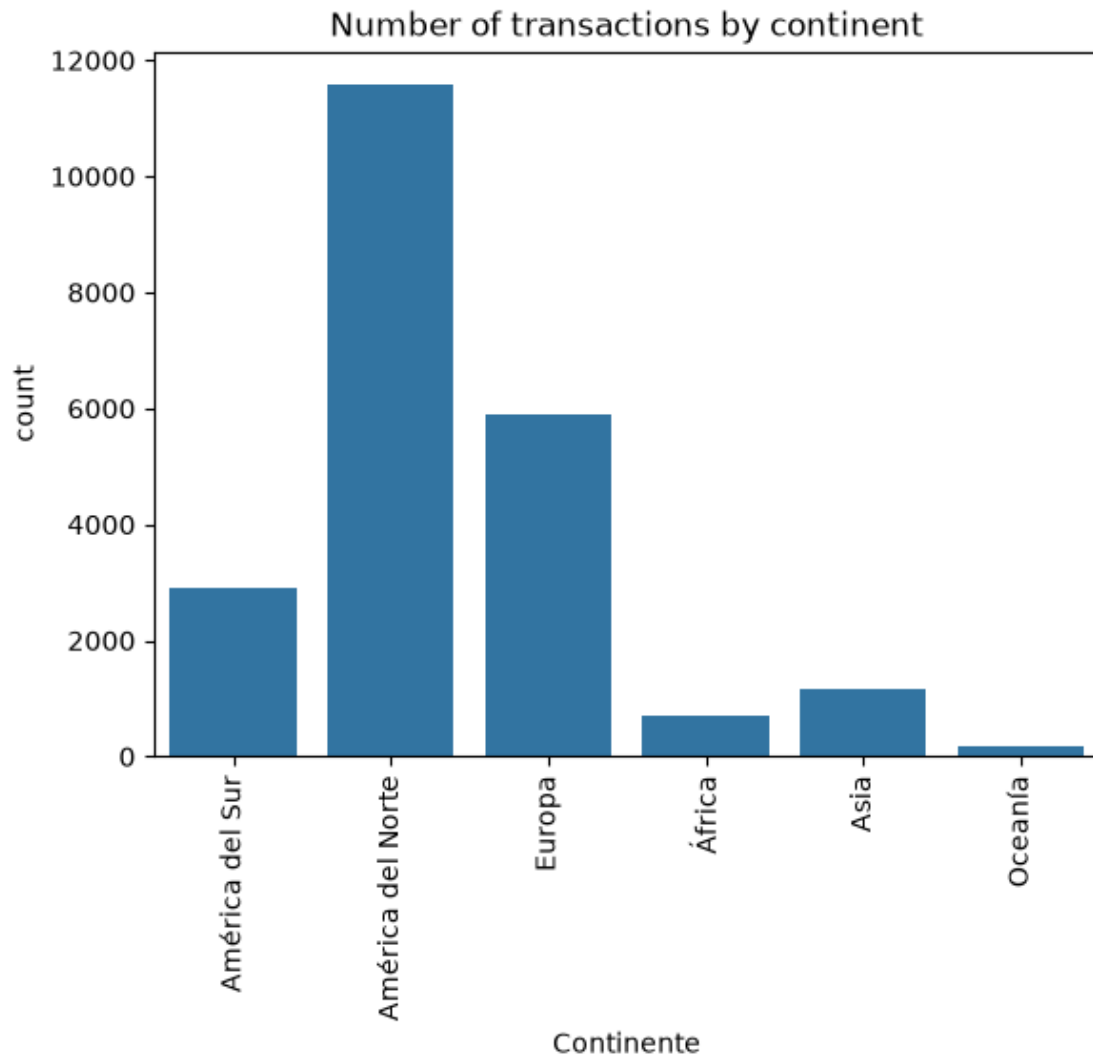
Observations: All features have 10 missing values, like the above table in data quality assessment showed

Number of transactions per continent

```
[85]: df_ventas[['País', 'Continente']] = df_ventas['Ubicación'].str.split(", ", expand=True)
```

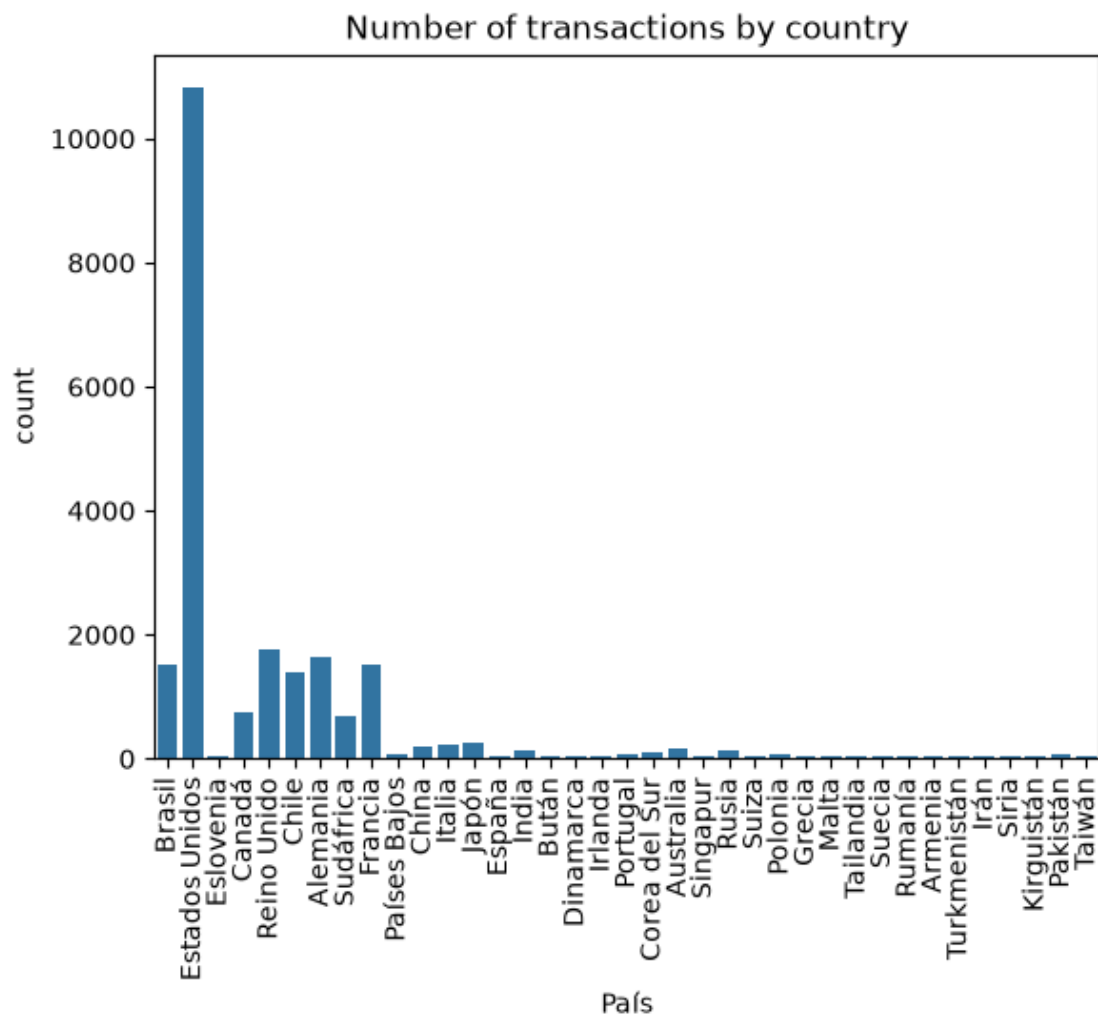
```
# drop missing values
df_ventas = df_ventas.dropna()
```

```
[86]: sns.countplot(df_ventas, x='Continente')
plt.title("Number of transactions by continent")
plt.xticks(rotation=90)
plt.show()
```



Number of sale transactions by country

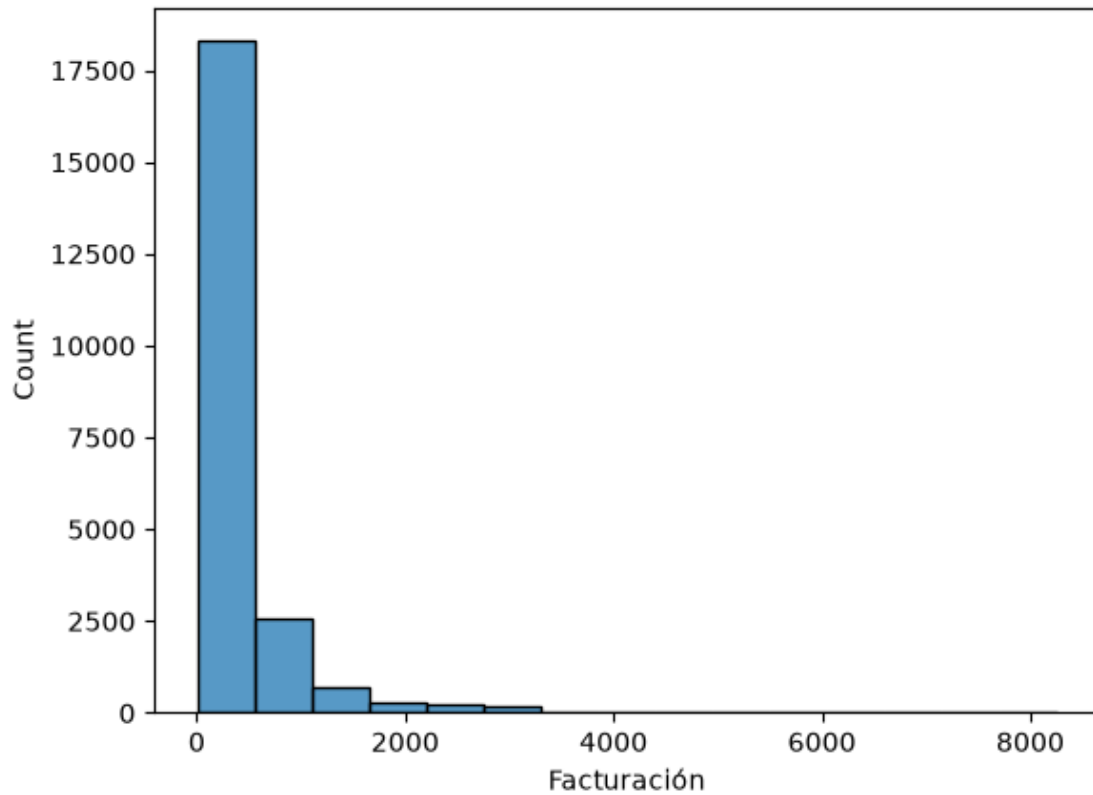
```
[87]: sns.countplot(df_ventas, x='País')  
plt.title("Number of transactions by country")  
plt.xticks(rotation=90)  
plt.show()
```



Revenue Distribution

```
[88]: sns.histplot(df_ventas, x="Facturación", bins=15)
plt.title("Revenue Distribution")
plt.title("")
```

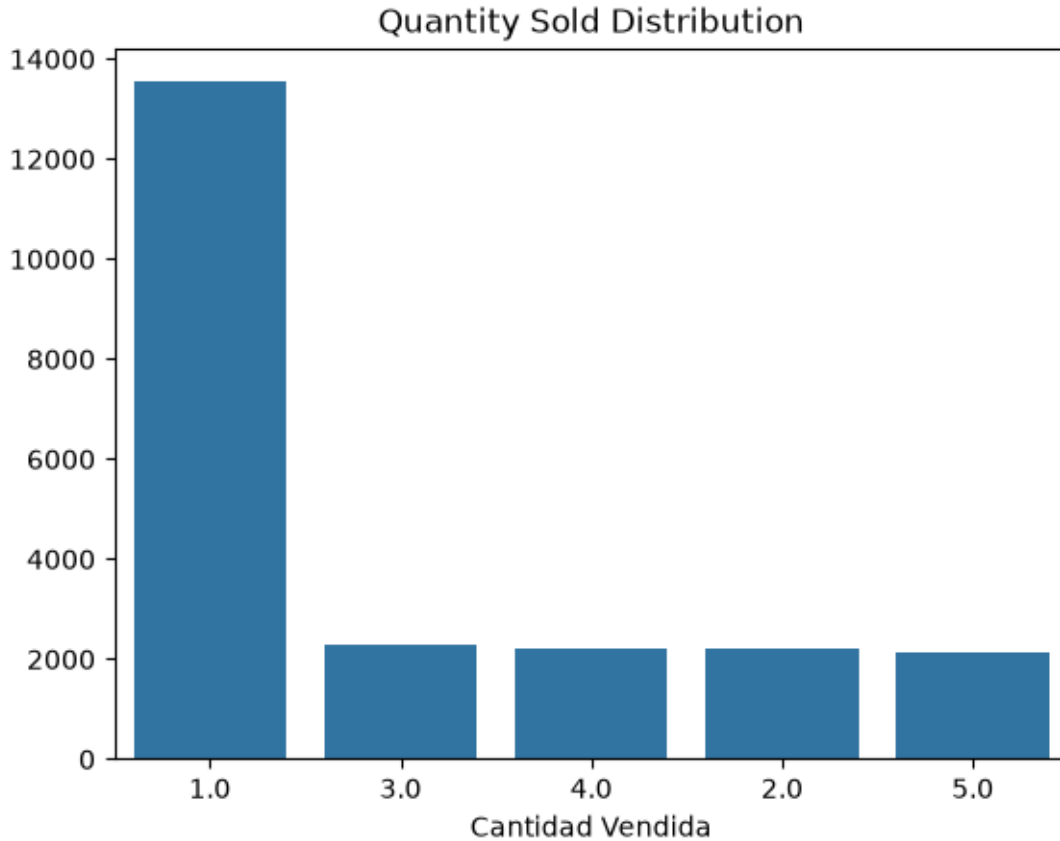
```
[88]: Text(0.5, 1.0, '')
```

Observation: The data in Facturación is highly right-skewed, showing that the most of the transactions have low revenue values, while a small number of transactions correspond to substantially higher revenue values. This type of pattern is common in retail dataset, which indicates the presence of high-value sales that can affect some aggregate metrics.

Sales quantity Distribution

```
[100]: quantity_sold = df_ventas["Cantidad Vendida"].value_counts()
sns.barplot(x=quantity_sold.index.astype("str"),
            y=quantity_sold.values)
plt.title("Quantity Sold Distribution")
plt.show()
```



Observation: The “Cantidad Vendida” feature is a discrete variable ranging between 1 and 5 units per transaction. Most transactions involve the purchase of only 1 unit, while transactions with more units sold occur less frequently. this can indicate that the bussines handles low-volume purchases.

1.1 EDA Conclusions

- The dataset contains 22,419 sales transactions across 36 countries, 7 brands, 20 product categories, and 444 unique products, providing sufficient coverage for commercial profitability analysis.
- Data quality assessment identified 10 completely empty records, which must be removed from the dataset. No significant issues were found regarding data types after validation.
- The Location field combined country and continent information in a single column. This attribute should be split into Country and Continent during the data preparation process to improve geographical analysis.
- The revenue distribution is highly right-skewed, indicating that most transactions generate relatively low revenue while a small number of transactions account for considerably higher sales values.
- The Quantity Sold variable shows that the majority of transactions involve the purchase of a single unit, suggesting that the business is primarily driven by low-volume purchases.
- Overall, the dataset demonstrates an adequate level of quality and consistency. After the

preprocessing stage, it is well prepared for the development of an executive dashboard focused on revenue, cost, gross profit, and gross profit margin analysis.
